discharge line to the meter should give a truly representative sample of the effluent. Sampling points should be arranged in all discharge pipes which have to be m~p'i~gfeg ~Pf ~ompl~ance w~th the Convention.
6.2.3 Where the Convention requires records, the oil content meter should be 'so designed and constructed that any operation carried out on them is

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SOURCE: MEPC.60(33).pdf
have to be m~p'i~gfeg ~Pf ~ompl~ance w~th the Convention.
6.2.3 Where the Convention requires records, the oil content meter should be 'so designed and constructed that any operation carried out on them is
automatically registered by the meter.
W/9146D/EWP
Resolution MEPC.60(33)
adopted on 30 October 1992
GUIDELINES AND SPECIFICATIONS FOR POLLUTION PREVENTION EQUIPMENT FOR MACHINERY SPACE BILGES OF SHIPSMEPC 33/20/Add.1
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ANNEX
The annex provides detailed Test and Performance Specifications for pollution prevention equipment and contains:
Part 1
Part 2
Part 3
Part 4
Part 5
Test and Performance Specifications for Type Approval of Oil Filtering Equipment
Test and Performance Specifications for Type Approval of Oil Content Meters for Bilge Alarms
Specifications for Environmental Testing for Type Approval of pollution prevention equipment
Method for the Determination of Oil Content
Documentation of Approval
PART 1 - TEST AND PERFORMANCE SPECIFICATIONS FOR TYPE APPROVAL OF OIL FILTERING EQUIPMENT
1.1 General
1.1.1 These Test and Performance Specifications for Type Approval relate

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SOURCE: MEPC.60(33).pdf
Documentation of Approval
PART 1 - TEST AND PERFORMANCE SPECIFICATIONS FOR TYPE APPROVAL OF OIL FILTERING EQUIPMENT
1.1 General
1.1.1 These Test and Performance Specifications for Type Approval relate
to oil filtering equipment. In addition, the electrical and electronic systems of the equipment should be tested in accordance with the Specifications for Environmental Testing contained in part 3 of this annex.
1.1.2 The equipment being tested should comply with the relevant requirements of the technical specifications contained in section 4.1 of these Guidelines

and Specifications.
1.2 Test Specifications
1.2.1 These Specifications relate to oil filtering equipment of low to medium capacity. Oil filtering equipment should be capable of giving an effluent containing not more than 15 ppm of oil irrespective of the oil content (from 0% to 100%) of the feed supplied to it.
1.2.2 The oil/water mixture, with which the system has in practice to deal, depends on:
.1 the position of the oil/water interface, with respect to the suction point, in the space being pumped;
.2 the type of pump used;
.3 the type and degree of closure of any control valve in the circuit; and

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depends on:
.1 the position of the oil/water interface, with respect to the suction point, in the space being pumped;
.2 the type of pump used;
.3 the type and degree of closure of any control valve in the circuit; and
.4 the general size and configuration of the system.
It is, therefore, desirable that the test rig be so constructed as to include not only the equipment, but also the pump and the most important of the valves, pipes, etc. (for an example see figures la and Ib). The pipework should be designed for a maximum liquid velocity of 3 m/s.
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Resolution MEPC.60(33)
adopted on 30 October 1992
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1.2.3 The tests should be carried out with a supply rate equal to the full throughput for which the equipment is designed.
1.2.4 Tests should be performed using two grades of oil. The tests described in 1.2.10 and 1.2.11 should be carried out using either:
. 1 a residual fuel oil (test oil All) of a relative density of not less than 0.94 at 15°C and of a viscosity of not less than

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SOURCE: MEPC.60(33).pdf
in 1.2.10 and 1.2.11 should be carried out using either:
. 1 a residual fuel oil (test oil All) of a relative density of not less than 0.94 at 15°C and of a viscosity of not less than 17 centistokes at 100°C (220 centistokes at 37.S0C), and a light distillate fuel oil (test oil all) having a relative density of not less than 0.83 at 15°C; or
.2 for ships which use residual fuel oil of higher density and viscosity than that referred to above, the test should be carried out with a residual fuel oil (test oil cl/) of relative density

not less than 0.9S at 15°C and a viscosity of not less than
25 centistokes at 100°C (440 centistokes at 37.8°C), and a light
distillate fuel oil (test oil all) having a relative density of
not less than 0.83 at 15°C.
If the equipment is fitted with heating facilities to allow the separated
oil retained in it to be discharged when the automatic discharge valve is
activated, the Certificate of Type Approval should be endorsed under the
heading limiting conditions imposed with the following statement:
"The equipment is fitted with heating facilities".
1.2.5 If the filtering equipment includes an integrated feed pump, this

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heading limiting conditions imposed with the following statement:
"The equipment is fitted with heating facilities".
1.2.5 If the filtering equipment includes an integrated feed pump, this
equipment should be tested with that pump supplying the required quantity of
oil and water to the equipment at its rated capacity. If the equipment is to
be fed by the ship's bilge pumps, then the unit will be tested by supplying
the required quantity of oil and water mixture to the inlet of a centrifugal
pump operating at not less than 1,000 rpm. This pump should have a delivery
capacity of not less than 1.5 times the rated capacity of the equipment at the
delivery pressure required for the test. The variation in oil/water ratio
will be obtained by valves on the oil and water suction pipes adjacent to the
pump suction, and the flow rate of oil and water or the oil content of the
supply to the equipment should be monitored. If a centrifugal pump is used,
the excess pump capacity should be dissipated by either a by-pass to the
suction side, or by a throttle valve or standard orifice plate on the
discharge side. In all cases, to ensure uniform conditions, the piping

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the excess pump capacity should be dissipated by either a by-pass to the
suction side, or by a throttle valve or standard orifice plate on the
discharge side. In all cases, to ensure uniform conditions, the piping
arrangements immediately prior to the equipment should be such that the
influent to the equipment should have a Reynolds Number of not less than
10,000 as calculated in fresh water, a liquid velocity of not less than 1 m/s
and the length of the supply pipe from the point of oil injection to the
equipment should have a length not less than 20 times its diameter. A mixture
inlet sampling point and a thermometer pocket should be provided near the
equipment inlet and an outlet sampling point and observation window should be
provided on the discharge pipe. Figures 1a and 1b give diagrammatic
representations of two possible test rigs, though it should be noted that the
water and oil from the equipment need not be led back to the supply tanks.
Where the water and oil are re-circulated during the test, additional sampling
points should be fitted in the water and oil lines to the mixture pump in

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water and oil from the equipment need not be led back to the supply tanks.

Where the water and oil are re-circulated during the test, additional sampling points should be fitted in the water and oil lines to the mixture pump in order to check the quality of the water and oil being supplied to the pump.

~I The reference is made to appendix for the Certificate of Type Approval for oil filtering equipment.

Figure 1 - Diagrammatic arrangements of test facilities

Typical test rigs:

Observat ion
window r-;::= =:t:--' .....-____ -, "I, -1:
Scum Oil
drain filtering I
Water
tlml:: ~---' ~ sample point
equipment Waler d r alO '----'"-_---'
Sample point
L. __ -